

- 4 The piston in the cylinder of a car engine is made to move in the cylinder with simple harmonic motion.

Fig. 4.1 shows the highest and lowest positions of the piston.

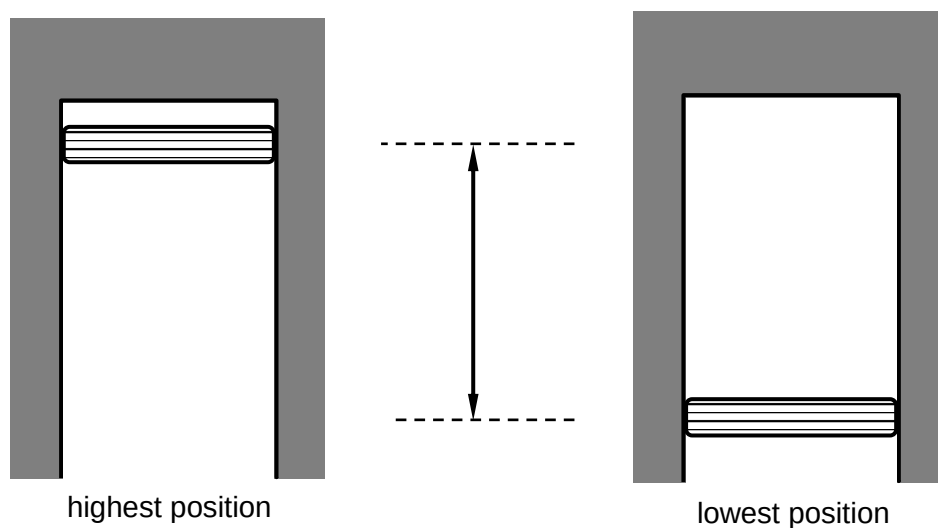


Fig. 4.1

The variation of the acceleration a of the piston with its displacement x from position O is as shown in Fig. 4.2.

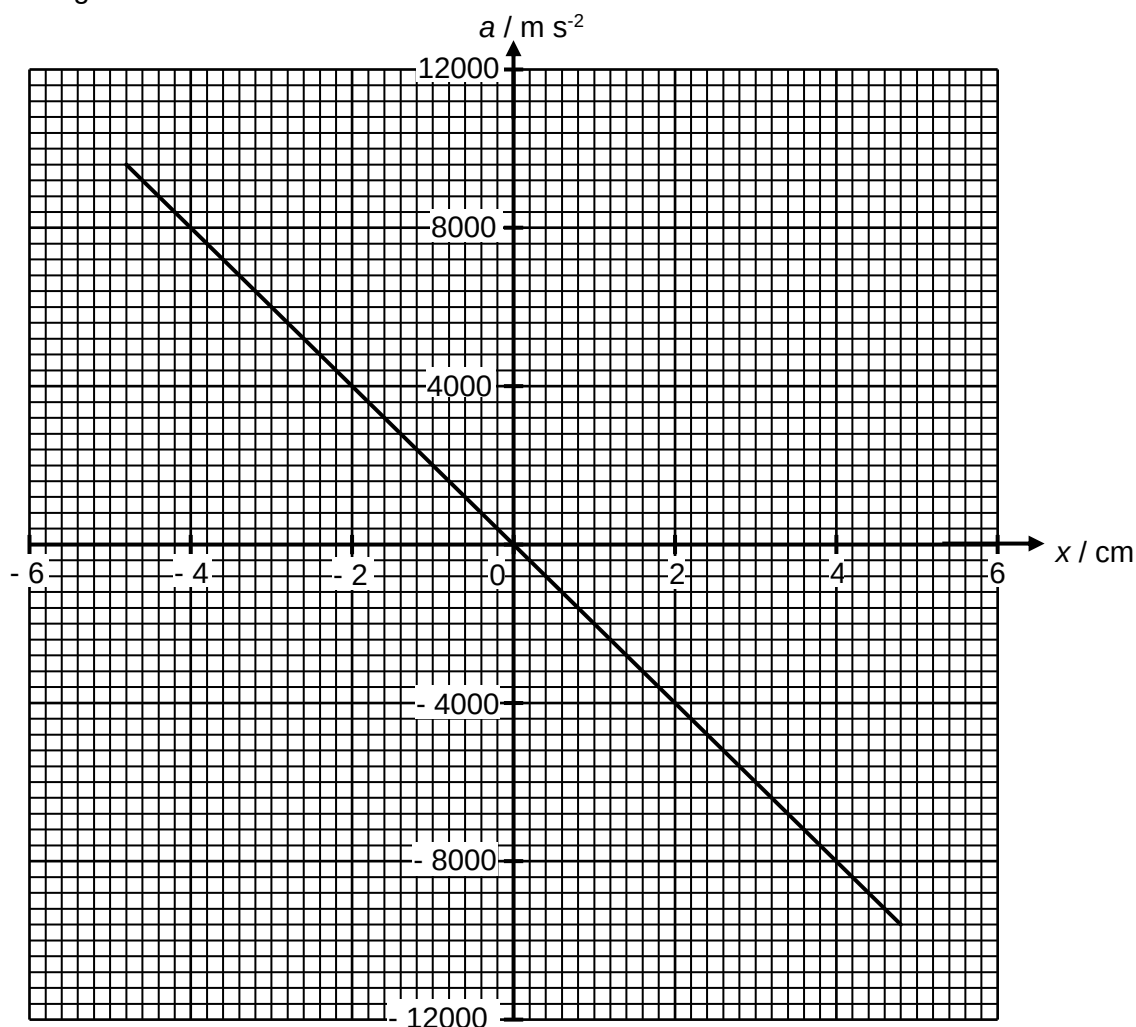


Fig. 4.2

- (a) State and explain the features of Fig. 4.2 that indicate that the motion of the piston is simple harmonic.

The graph is a straight line passing through origin implying the acceleration is

proportional to the displacement from a fixed point. [B1]

The negative gradient indicate that the direction of acceleration is always opposite to the displacement. [B1]

[2]

- (b) Determine the maximum speed of the piston.

$$\text{gradient} = -\omega^2$$

$$\frac{9600 - 0}{4.8 \times 10^{-2}} = -\omega^2 \quad [\text{B1}]$$

$$v_o = \omega x_o = 447(4.8 \times 10^{-2}) = 21 \text{ m s}^{-1} \quad [\text{B1}]$$

maximum speed = m s⁻¹ [2]

- (c) With reference to Fig. 4.2,

- (i) explain why the time taken for the piston to move from $x = 3.3 \text{ cm}$ to $x = 0 \text{ cm}$ is the same as that from $x = 0 \text{ cm}$ to $x = -3.3 \text{ cm}$.

The magnitude of the acceleration is symmetrical about origin. [B1] This results in the speed to be the same value at the distance from origin as the velocity at the origin is the maximum. Hence the time taken for the piston to move from $x = 3.3 \text{ cm}$ to origin and from origin to $x = -3.3 \text{ cm}$ is the same as the distance travelled is the same. [B1]

[2]

- (ii) The area under the graph in Fig. 4.2 from $x = 0$ to $x = 4.8$ cm is given by Z .

A student calculates K , the maximum kinetic energy of the piston, using the relationship

$$K = Z \times M$$

where M is the mass of the piston.

Explain why this relationship is valid.

By N2L, the product of mass of piston and the acceleration of piston represent the net force acting on the piston. [B1]
 The area under the graph for the variation of net force with displacement represent the work done on the system to start the oscillation. Since the work done on the system results in the piston gaining a maximum kinetic energy when it is at the origin, the area under the graph $\times$ mass = maximum kinetic energy piston. [B1]
[2]

- (d) The piston is made to move by connecting a rod to a rotating crankshaft as shown in Fig. 4.3. As the pivot P on the crankshaft rotates, the piston will move up and down.

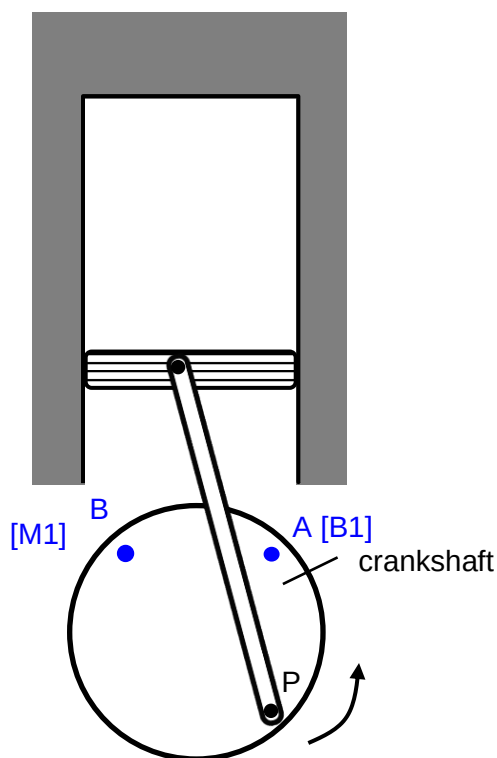


Fig. 4.3

When pivot P is at the position shown in Fig. 4.3, the piston is moving upward with a position of $x = -3.3$ cm.

Indicate on Fig. 4.3 the position of pivot P when the piston is at the following positions,

- (i) $x = 3.3$ cm moving upwards. (Indicate this position with A)
 (ii) $x = 3.3$ cm moving downwards. (indicate this position with B) [2]

[Total: 10]

5 An isolated spherical conductor has charge q as shown in Fig 5.1